

Data Collection and Preprocessing Phase

Date	21 April 2026
Name	Srishti Sanjay More
Project Title	Restaurant Recommendation System
Maximum Marks	6 Marks

Data Preprocessing

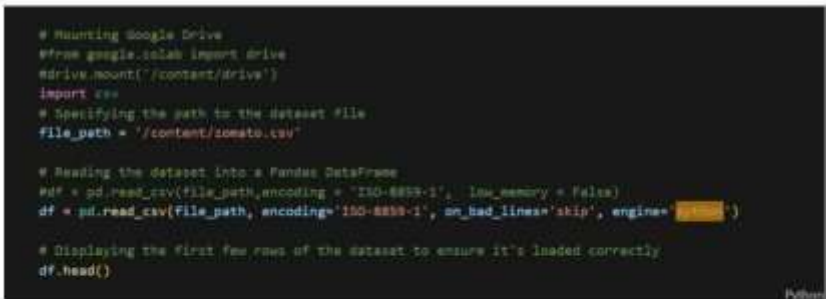
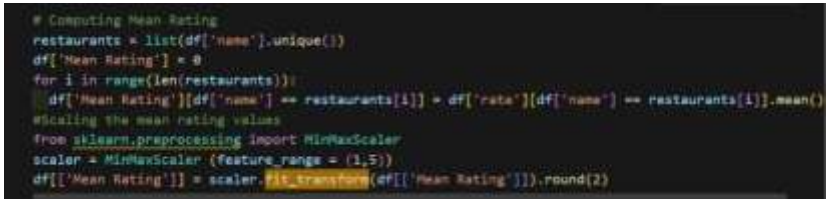
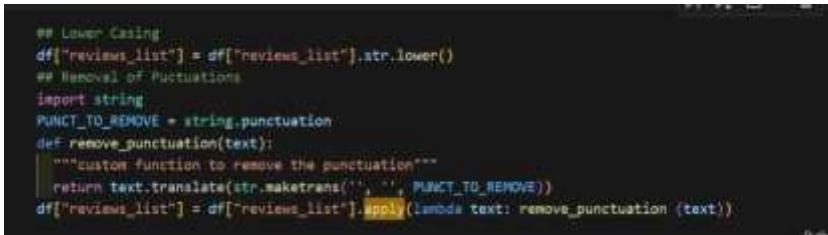
The images will be preprocessed by resizing, normalizing, augmenting, denoising, adjusting contrast, detecting edges, converting color space, cropping, batch normalizing, and whitening data. These steps will enhance data quality, promote model generalization, and improve convergence during neural network training, ensuring robust and efficient performance across various computer vision tasks.

Section	Description
Data Overview	The dataset contains restaurant information from Zomato, including name, reviews, ratings, cuisines, cost, and more. The data is cleaned, deduplicated, and preprocessed for building a content-based recommendation system.
Resizing	<i>Not applicable for text data.</i>

Normalization	Ratings are normalized to a 1-5 scale using MinMaxScaler. Text is lowercased and punctuation is removed.
Data Augmentation	Not applicable for text data.
Denoising	Text is cleaned by removing newline characters and punctuation.
Edge Detection	Not applicable for text data.

Color Space Conversion	Not applicable for text data.
Image Cropping	Not applicable for text data.
Batch Normalization	Not applicable for text data.

Data Preprocessing Code Screenshots

Loading Data	 <pre> # Mounting Google Drive from google.colab import drive drive.mount('/content/drive') import csv # Specifying the path to the dataset file file_path = '/content/dataset.csv' # Reading the dataset into a Pandas DataFrame df = pd.read_csv(file_path, encoding = 'ISO-8859-1', low_memory = False) df = pd.read_csv(file_path, encoding='ISO-8859-1', on_bad_lines='skip', engine='python') # Displaying the first few rows of the dataset to ensure it's loaded correctly df.head() </pre>
Resizing	Not applicable
Normalization	 <pre> # Computing Mean Rating restaurants = list(df['name'].unique()) df['Mean Rating'] = 0 for i in range(len(restaurants)): df['Mean Rating'][df['name'] == restaurants[i]] = df['rate'][df['name'] == restaurants[i]].mean() # Scaling the mean rating values from sklearn.preprocessing import MinMaxScaler scaler = MinMaxScaler(feature_range = (1,5)) df[['Mean Rating']] = scaler.fit_transform(df[['Mean Rating']]).round(2) </pre>
Data Augmentation	Not applicable
Denoising	 <pre> ## Lower Casing df["reviews_list"] = df["reviews_list"].str.lower() ## Removal of Punctuations import string PUNCT_TO_REMOVE = string.punctuation def remove_punctuation(text): """Custom function to remove the punctuation""" return text.translate(str.maketrans('', '', PUNCT_TO_REMOVE)) df["reviews_list"] = df["reviews_list"].apply(lambda text: remove_punctuation(text)) </pre>
Edge Detection	Not applicable

Color Space Conversion	<i>Not applicable</i>
Image Cropping	<i>Not applicable</i>
Batch Normalization	<i>Not applicable</i>